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Interpretable Multimodal Cancer Prototyping with Whole Slide Images and Incompletely Paired Genomics

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***Abstract*— Multimodal approaches that integrate histology and genomics hold strong potential for precision oncology. However, phenotypic and genotypic heterogeneity limits the quality of intra-modal representations and hinders effective inter-modal integration. Furthermore, most existing methods overlook real-world clinical scenarios where genomics may be partially missing or entirely unavailable. We propose a flexible multimodal prototyping framework to integrate whole slide images and incomplete genomics for precision oncology. Our approach has four key components: 1) Biological Prototyping using text prompting and prototype-wise weighting; 2) Multiview Alignment through sample- and distribution-wise alignments; 3) Bipartite Fusion to capture both shared and modality-specific information for multimodal fusion; and 4) Semantic Genomics Imputation to handle missing data. Extensive experiments demonstrate the consistent superiority of the proposed method compared to other state-of-the-art approaches on multiple downstream tasks. The code is available at GitHub.**

***Index Terms*— Multimodal Learning, Prototype Learning, Computational Pathology, Missing Modality**

I. Introduction

PRECISION oncology increasingly relies on multimodal approaches that provide complementary information. Whole slide images (WSIs) capture tissue morphology and cellular phenotypes, whereas genomics probes underlying molecular mechanisms [1]. Integrating both offers interpretability and improvements in cancer diagnosis, prognosis, and treatment stratification with phenotypic and genotypic characteristics [2]–[4].

Due to the high dimensionality and inherent disparity of WSIs and genomics, challenges remain for effective multimodal learning: **1) Suboptimal WSI representation:** WSIs are gigapixel-scale images that typically lack detailed patch-level annotations [5], resulting in limited label or semantic information for representation learning. To model WSI, most existing methods employ multiple instance learning (MIL), where patch-level embeddings are aggregated into slide-level. However, these methods treat patches as independent instances [6], thereby ignoring key cancer phenotypes, such as spatial tissue structures that reflect tumor microarchitecture and cellular interactions. Recent advances have introduced attention-based MIL [7] that models patch dependencies, aiming to focus on the most relevant regions. In parallel, graph-based WSI methods [8], [9] explicitly model inter-patch relationships by representing patches as nodes and their spatial or morphological affinities as edges, improving modeling of tissue architecture and local context. However, these approaches are still predominantly optimized with slide-level supervision, leaving patch-level semantics weakly exploited. As a result, the learned representations may emphasize non-diagnostic or texture-biased regions rather than pathology-relevant patterns [10], [11], leading to suboptimal WSI representations that may fail to fully capture clinically meaningful features. Most recent methods [3], [12], [13] aggregate patches into prototypes, leveraging morphological redundancy in histology to construct generalized WSI representations. However, the resulting prototype clusters often require expert annotation to be mapped to pathological concepts, limiting their clinical utilization.

2) Ineffective cross-modal alignment due to semantic gaps: Histology and genomics describe biological processes at distinct levels, resulting in substantial semantic discrepancies between feature spaces. Aligning distinct modalities before integration is crucial for multimodal learning [14]. Contrastive learning shows promise for cross-modal alignment [15], encouraging paired WSI-genomic representations to converge while separating unpaired ones [16]. However, in most clinical datasets, patients have multiple WSIs but only a single genomic profile [17]. This imperfect pairing introduces ambiguity and noise in alignment, limiting the discriminative

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power of learned representations. Constructing robust and coherent cross-modal embeddings remains a challenge.

3) Coarse integration strategies that overlook modality-specific features: Effectively integrating WSIs and genomics requires not only capturing their shared biological signals (e.g., proliferation or metabolics), but also preserving modality-specific signals (e.g., spatial context in histology or mutations in genomics) [13]. Most existing approaches either embed both modalities into a unified latent space or focus solely on modeling global correlations [18], [19]. While this facilitates fusion, it often suppresses modality-unique information and limits interpretability. To harness the full potential of cross-modal complementarity, a fine-grained integration is required. However, achieving this balance is non-trivial as it requires disentangling modality-specific signals from shared representations while minimizing redundancy.

4) Limited robustness under incomplete genomics: In real-world clinical scenarios, genomic data are often incomplete, inconsistently acquired, or entirely absent due to time constraints, tissue availability, or sequencing costs [20]. Most existing methods [4], [21] typically assume complete data availability across modalities or distill histology-only models to address incompleteness [20]. The lack of flexibility in handling heterogeneously available genomics, partially or entirely missing genomics, limits their generalizability and clinical deployment. More recently, a few studies have emerged to address incomplete multimodal survival prediction using generative [22], distillation-based [23], [24], and retrieval-based [25], [26] strategies. However, how to ensure biological consistency of generated genomics is still an open question.

To address these challenges, we propose a multimodal learning framework integrating WSIs and incompletely paired genomics. **Firstly**, a **Biological Prototyping** scheme is proposed to model interpretable intra-modal representations. For WSIs, we leverage pathology domain-informed text prompts to derive meaningful visual histology prototypes from unannotated WSI patches. For genomics, we introduce functional grouping based on known biological functions to construct genomic prototypes. A prototype-wise weighting mechanism is further proposed to assign importance scores to prototypes, enhancing explainability and allowing models to focus on key modality-specific features. **Secondly**, a **Multiview Alignment** strategy is proposed to bridge semantic gaps across modalities. We first introduce a mutual information-motivated loss to ensure distribution-level alignment and global semantic consistency. To alleviate imperfect pairing in real-world multimodal datasets, we further propose a sample-wise alignment that captures inter-sample relational consistency across mini-batches. Together, these strategies facilitate robust and complementary geno-phenotypic representations. **Thirdly**, we propose a **Bipartite Fusion** that captures both shared and unique features across modalities by leveraging cross-modal affinity, enabling fine-grained semantic fusion while maintaining modality-specific information, thereby enhancing both performance and interpretability. **Finally**, to improve robustness in the presence of missing genomics, we design a **Semantic Genomics Imputation** module. This module flexibly handles various missingness scenarios, including fully missing, partially missing, or absent during inference, thereby enhancing clinical applicability. Extensive experiments on multiple cancer downstream tasks in three datasets demonstrate that our method outperforms other state-of-the-art (SOTA) methods.

II. Related Work

A. Multimodal Learning for Precision Oncology

Recent multimodal approaches with WSIs and genomics primarily focus on two aspects: intra-modal representations and inter-modal integrations. For WSIs, traditional approaches employed MIL to aggregate patch-level features into slide-level representations. However, these methods typically neglect inter-patch correlations and lack biological interpretability, limiting their clinical utility. For genomics, recent studies [4], [21], [27] employ a coarse categorization into six genomics families; others [3], [28] utilize biological pathways for transcriptomic representation. Different from these studies, we propose a biological prototyping scheme that simultaneously achieves two critical objectives: (1) capturing biologically relevant patterns in WSI patches and genomic tokens to form modality-specific prototypes, and (2) dynamically quantifying prototype importance through predicted weights to enable model interpretation at both phenotype and genotype levels.

To integrate histology and genomics, earlier studies integrated WSI Region Of Interests (ROI) and genomics profiles [2], [20], [29]. Although straightforward, ROIs are not readily available in real-world clinical scenarios. Recent studies have focused on WSIs and transcriptomes [3], [28]. For instance, SurvPath [28] introduced biological pathways for transcriptomic modelling and integrated WSIs to predict patient survival. However, the inherent semantic gap between genotype and phenotype hinders multimodal fusion, ultimately degrading model performance. Unlike conventional approaches, our method employs: **(i)** Multiview Alignment to reduce semantic discrepancies through cross-modal correlation learning, and **(ii)** a Bipartite Fusion module that dynamically enhances discriminative features while suppressing noise, enabling fine-grained prototypes with enriched semantic representations.

Most recent multimodal methods have moved beyond simple fusion toward more structured representation learning. A multimodal prototyping was introduced in MMP [3] for cancer survival prediction, demonstrating the value of prototype-based cross-modal abstraction. In PIBD [13] prototype learning was further combined with modality disentanglement to separate shared and modality-specific information. These methods are closely related to our work, but their aggregated prototypes typically rely on expert annotation to be aligned with pathological concepts, and such post hoc interpretability may limit their clinical use.

B. Foundational Multimodal Pretraining

Recent pathology vision-language and multimodal pretraining methods have shown that large-scale pretraining can provide semantically structured representation spaces for pathological data [30]. For example, PLIP [31] performed contrastive language-image pretraining on paired pathology images and natural language descriptions, while CONCH [32],

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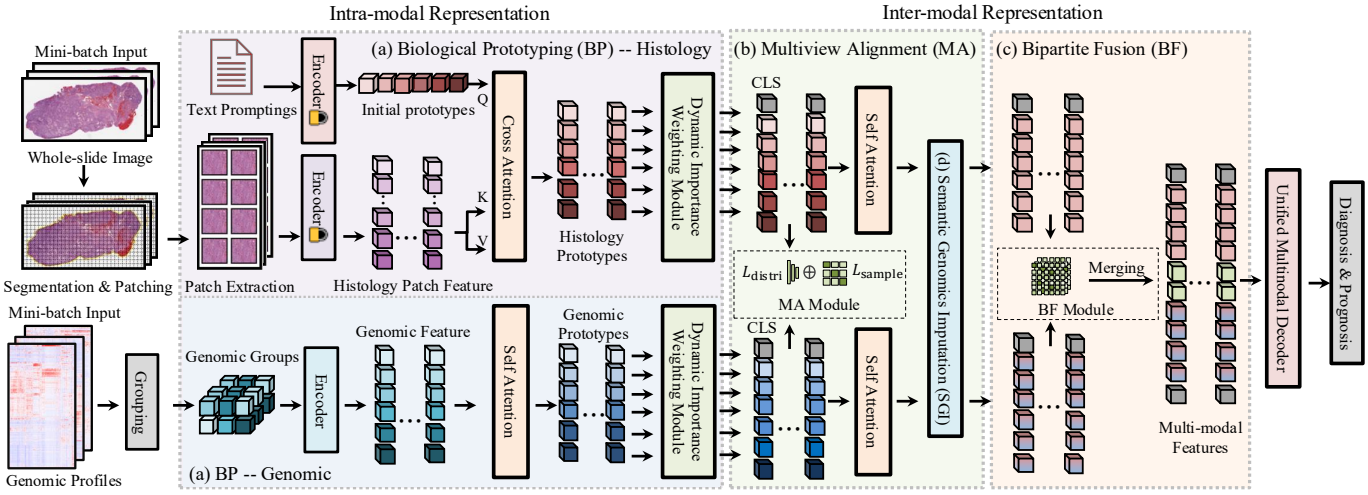


Fig. 1. Overview of the proposed framework. (a) Biological Prototyping (BP): extracts biologically meaningful prototypes from WSI using class-specific prompting and dynamic importance weighting, while genomic prototypes are driven from functional groups. (b) Multiview Alignment (MA): bridges the semantic gap between WSI and genomic features via sample-wise alignment and mutual information-motivated cross-modal correlation learning. (c) Bipartite Fusion (BF): integrates shared and modality-specific features for multimodal fusion. (d) Semantic Genomics Imputation (SGI): imputes missing genomic features directly in the feature space.

PathChat [33] and MUSK [34] further extended this line with larger-scale vision-language or multimodal pretraining. These advances are relevant to our work because they provide pathology-aware semantic priors for representation learning. In particular, we leverage the pretrained PLIP [31] text encoder as a pathology-specific feature extractor to encode histology prompts for prototype initialization. This enables our Biological Prototyping module to start from semantically grounded pathology concepts, which are subsequently refined by cross-attention with WSI patch features and further integrated with genomics through multiview alignment, bipartite fusion, and semantic genomics imputation.

C. Multimodal Learning with Incomplete Modality

The absence of genomic data poses a critical challenge to multimodal learning, as acquiring comprehensive genomics profiles remains expensive and relies on tissues [29], [35]. To tackle this, Xing et al. [20] transferred multimodal teacher knowledge to a uni-modal student model by a distillation framework for glioma grading. Wang et al. [36] designed a multi-task framework that jointly predicted molecular status and glioma classification from WSIs, requiring genomics only during training. Despite promising, these methods still rely on paired multimodal data during training. Qiu et al. [37] proposed an imputation-based approach, enabling the use of incomplete histology and genomics during training. However, the substantial domain gap between histology and genomics limits the effectiveness of direct imputation. Recent studies have emerged to address missing modalities from different perspectives. LD-CVAE [22] adopted a generative variational framework for incomplete multimodal survival modeling, while distillation- and prompt-based methods [23] explored missing-information prediction through knowledge transfer. M^2Surv [26] and M^3Surv [25] further introduced the retrieval-based frameworks for robust survival prediction from multi-slides and multi-omics data. In contrast, our SGI module focuses on semantic genomics imputation guided by aligned multimodal representations within a framework of interpretable prototype-based histology-genomics, enabling our model to adapt to various real-world scenarios.

III. Methodology

Fig. 1 provides an overview of the proposed multi-modal prototype learning approach designed to model intra- and inter-modal representations under WSIs and incompletely paired genomics. The proposed framework includes: 1) a prompting-based Biological Prototyping module for interpretable intra-modal representation (Section III-A); 2) a Multi-View Alignment module (Section III-B) and Bipartite Fusion (Section III-D) for inter-modal representation; 3) a Semantic Imputation module to handling incomplete genomics (Section III-C). Section III-E shows the algorithm workflow.

A. Biological Prototyping

To construct biologically meaningful and interpretable intra-modal representations, we propose the Biological Prototyping (BP) module (Fig. 1(a)). BP initializes histology prototypes using pathology-specific text prompts (Section III-A.1), and genomics are divided into prototypes with established functional groups (Section III-A.2), grounding both modalities in meaningful biological concepts. These prototypes serve as semantic anchors that guide feature learning toward recognizable histological and molecular patterns. A dynamic importance weighting module further quantifies each prototype's contribution to downstream tasks, enhancing transparency.

1) Histology prototypes: We define six coarse-grained histological categories: *Neoplastic, Necrotic, Inflammatory, Stromal, Infiltrative*, and *Other Cell Types*, following prior work [38], [39]. Each category is described by multiple pathology-informed text prompts to provide biologically meaningful semantic priors. Initial prototype embeddings for

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each category are obtained using the text encoder of PLIP [31], which is pretrained on pathology image-text pairs. We use PLIP here as a pathology-specific feature extractor for histology prompts, providing semantically grounded initialization for the histology prototypes before their refinement through cross-attention with WSI patch features. To adapt prototypes to tumor heterogeneity and contextual variations within WSIs, we refine them using a cross-attention mechanism where patch embeddings act as keys/values and initial prototypes as semantic queries:

$$P^t = \text{softmax}\left(\frac{P^{t-1}W_q\left(P_{\text{patch}}W_k\right)^T}{\sqrt{D}}\right)\left(P_{\text{patch}}W_v\right) \quad (1)$$

where P^t denotes updated prototypes at iteration t-th and P^{t-1} represents histology prototypes at the $(t-1)$-th interation. This mechanism encourages prototypes to evolve into semantically meaningful cluster centers.

2) Genomic prototypes: As genes already have functional groups with established biological definitions, we do not use text-prompt initialization for genomic prototypes; instead, following established molecular taxonomies [21], genes are grouped by their functional roles: *Tumor Suppressor Genes, Oncogenes, Protein Kinases, Cell Differentiation Markers, Transcription Factors*, and *Cytokines and Growth Factors*. These grouped genes are encoded by a trainable SNN-based encoder [40] to form six genomic prototypes, with functionally coherent information encoded in each prototype. These prototypes are refined via a self-attention mechanism, modeling genotype-genotype dependencies. This allows genomic prototypes to adaptively represent salient molecular patterns.

3) Dynamic importance weighting: To explicitly compute the contribution of each prototype toward downstream tasks, we introduce task-specific importance score w for each prototype:

$$w_i = \sigma\left(f_\theta\left(z_i\right)\right) \quad (2)$$

where σ, f_θ, and z_i denote the sigmoid function, linear layers, and i-th prototype token, respectively. This mechanism allows the model to highlight informative features while suppressing irrelevant ones, improving interpretability and stability.

B. Multiview Alignment

WSIs and genomics inherently exhibit semantic gaps. Multimodal cancer datasets often contain imperfectly paired samples, which exacerbates the modality discrepancy. To mitigate these challenges, we propose the Multiview Alignment (MA) module as shown in Fig. 1(b) and Fig. 2, which aligns the two modalities at both distribution-wise and sample-wise levels. Specifically, distribution-wise alignment enhances cross-modal coherence through a mutual information (MI)-motivated correlation objective, while the sample-wise strategy calibrates alignment by cross-sample batch consistency. Jointly optimizing these two alignment strategies effectively bridges modality gaps before the final fusion.

1) Distribution-wise Alignment via Cross-modal Correlation: MI quantifies the shared information between two modalities, serving as a robust measure of cross-modal dependency. As illustrated in Fig. 2, we optimize cross-modal correlation

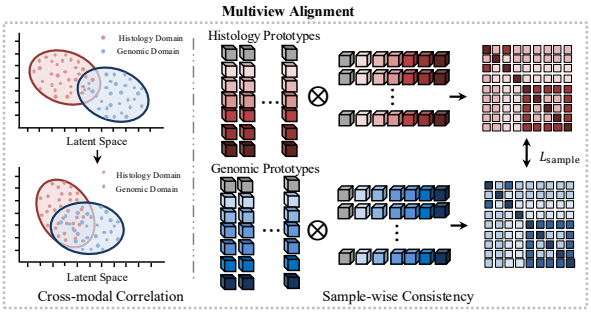


Fig. 2. Multiview Alignment (MA): Bridges the semantic gap between WSI and genomic features using mutual information-motivated cross-modal correlation learning and sample-wise alignment. Left: Maximize cross-modal correlation in MA. Right: Sample-wise consistency.

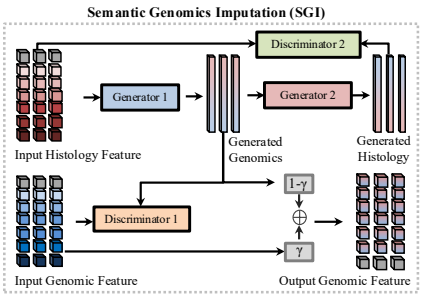


Fig. 3. Semantic Genomics Imputation (SGI): A CycleGAN-based network to generate genomic features from histology features, with a progressive interpolation to ensure robustness in various settings.

to enforce distributional alignment between histology and genomic prototypes. For a mini-batch paired histology $P \in \mathbb{R}^{B\times N_P\times D_P}$ and genomic embeddings $G \in \mathbb{R}^{B\times N_G\times D_G}$ (B as the batch-size, N as the number of prototypes, D as the feature dimension), MI between modalities is defined as $I(P,G) = H(P) - H(P|G)$, $H(P)$ and $H(P|G)$ denote the entropy and conditional entropy. Directly computing MI is challenging due to the intractability of high-dimensional probability distributions. We therefore operationalize this objective using a cross-modal correlation surrogate inspired by canonical correlation analysis (CCA). Given feature-wise standardized representations $\widetilde{P}$ and $\widetilde{G}$, their cross-correlation matrix C_{PG} is computed as $\frac{1}{B-1}\widetilde{P}^\top\widetilde{G}$. We then perform singular value decomposition:

$$C_{PG} = U\,\text{diag}(\sigma_1,\ldots,\sigma_r)V^\top, \quad (3)$$

where σ_k denotes the k-th singular value and $r = \min(D_P, D_G)$. The distribution-wise alignment loss is

$$L_{\text{distri}} = \left(\frac{1}{r}\sum_{k=1}^{r}\sigma_k + \epsilon\right)^{-1}. \quad (4)$$

Minimizing this loss maximizes the overall cross-modal correlation. For jointly Gaussian and whitened representations, these singular values correspond to canonical correlations, whose maximization increases a lower bound on MI [41], [42]. Therefore, we use this CCA-inspired objective as a practical surrogate for MI-motivated distribution-wise alignment.

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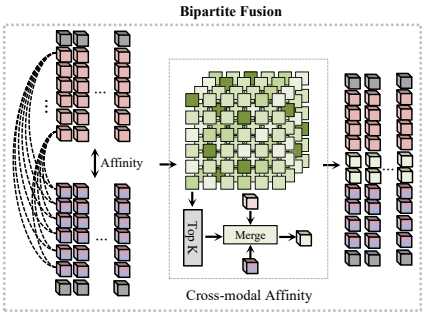


Fig. 4. Bipartite Fusion: Utilizing prototype-wise affinity to integrate shared and modality-specific features for multimodal fusion.

2) Sample-wise Alignment via Correlation Consistency: Conventional contrastive-based alignment approaches treat each WSI-genomic pair as an independent positive sample, inadvertently pushing WSIs from the same patient apart as negatives, leading to suboptimal representations. To mitigate this, we align relational structures across modalities rather than individual embeddings. Given P and G as histology and genomic feature matrices within a batch of B samples, we first apply L2-normalization to obtain scale-invariant representations: $\tilde{P}$ and $\tilde{G}$. We then compute their Gram matrices to capture intra-batch correlations:

$$M_P = \tilde{P}\tilde{P}^\top, \quad M_G = \tilde{G}\tilde{G}^\top, \tag{5}$$

where $M_P(i,j)$ and $M_G(i,j)$ represent the similarity between samples i and j within the histology and genomic modalities, respectively. If two samples exhibit strong similarity in the genomic space, their corresponding WSIs should display comparable similarity patterns in the histology space. We therefore enforce relational consistency by minimizing the Frobenius norm between M_P and M_G:

$$L_{\text{sample}} = \frac{1}{B^2}\|M_P - M_G\|_F^2, \tag{6}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. By aligning cross-modal representations at a structural level rather than individual embeddings, this design preserves intra-patient and inter-sample relationships across modalities, preserving biological coherence under imperfect WSI-genomic pairing.

Finally, the MA module combines both sample-wise and distribution-wise alignment losses:

$$L_{\text{MA}} = L_{\text{sample}} + L_{\text{distri}} \tag{7}$$

To capture inter-prototype interactions after alignment, we introduce a learnable $[CLS]$ token and apply a self-attention layer to each modality, enabling the model to encode dependencies and produce a global contextual representation.

C. Semantic Genomics Imputation

To handle incomplete genomic data in clinical practice, we propose a Semantic Genomics Imputation (SGI) module (Fig. 3) that infers missing genomic embeddings directly from histology features. Benefiting from the pre-aligned feature space with MA, the imputed genomic features maintain cross-modal consistency, supporting robust predictions when genomics are absent.

Inspired by the cycle-consistency in CycleGAN [43], SGI performs bidirectional feature-level translation between histology and genomic representations: $F_{P\to G}$ translates histology embeddings into genomic embeddings, and $F_{G\to P}$ reconstructs histology embeddings from genomic features. The forward cycle consistency loss is defined as:

$$L^P_{\text{cycle}} = E_{p_i\sim P}\left\|F_{G\to P}\left(F_{P\to G}\left(p_i\right)\right) - p_i\right\|_1 \tag{8}$$

enforcing reconstruction of the original histology embedding after bidirectional translation. A symmetric term $L\text{cycle}^G$ is similarly applied for genomic embeddings under reverse transformation.

Beyond reconstructive consistency, we encourage imputed genomic embeddings to follow the true statistical distribution of observed ones. Thus, we incorporate an adversarial regularization strategy, where discriminators D_G and D_P are trained to distinguish between real and generated features in genomic and histology embedding spaces, respectively. The adversarial loss for histology-to-genomics translation is defined as:

$$L^G_{\text{adv}} = E_{g_i\sim G}\left[logD_G\left(g_i\right)\right] + E_{p_i\sim P}\left[log\left(1 - D_G\left(F_{P\to G}\left(p_i\right)\right)\right)\right] \tag{9}$$

A symmetric adversarial loss L^P_{adv} is defined analogously for genomics-to-histology translation. The final optimization objective integrates cycle consistency and adversarial learning:

$$L_{\text{SGI}} = L^G_{\text{adv}} + L^P_{\text{adv}} + \lambda\left(L^G_{\text{cycle}} + L^P_{\text{cycle}}\right) \tag{10}$$

where λ controls the relative importance of cycle consistency loss and adversarial loss.

To stabilize training under varying levels of missingness, we adopt a progressive interpolation strategy that gradually shifts reliance from real to generated genomic embeddings:

$$\tilde{g_i} = m_i \cdot g_i + (1 - m_i) \cdot \hat{g_i} \tag{11}$$

where g_i is a real genomic embedding, $\hat{g_i}$ is the generated one, and m_i decreases over iterations. This smooth transition enhances robustness and allows the model to adapt seamlessly to different degrees of missing genomic data.

D. Bipartite Fusion

To jointly preserve modality-specific semantics and capture cross-modal interactions, we propose a Bipartite Fusion (BF) module (Fig. 4). In this design, histology and genomic prototypes form two disjoint node sets in a bipartite graph, where edges encode their pairwise affinities. Given the i-th paired histology and genomic prototype sets $p_i \in \mathbb{R}^{N_P\times D_P}$ and $g_i \in \mathbb{R}^{N_G\times D_G}$, the prototypes affinity is computed as:

$$A_i(n,m) = \frac{p_{i,n}^\top g_{i,m}}{\|p_{i,n}\|\,\|g_{i,m}\|}, \tag{12}$$

where $A_i \in \mathbb{R}^{N_P\times N_G}$ captures the affinity between the n-th histology prototype and the m-th genomic prototype.

We select the Top-K most affine cross-modal prototype pairs (p_{i,n_t}, g_{i,m_t}) to perform cross-modal fusion, enabling localized and interpretable interactions between modalities. For each selected pair, the fused prototype is obtained by element-wise averaging:

$$f_{i,t} = \frac{p_{i,n_t} + g_{i,m_t}}{2}, \quad t = 1,\ldots,K, \tag{13}$$

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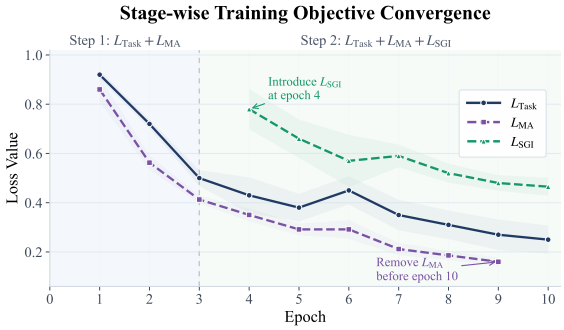


Fig. 5. Alternating Training Strategy consists of two phases for each downstream task: step 1 for completed multimodal learning ($\boldsymbol{L_{\text{Task}}}$ and $\boldsymbol{L_{\text{MA}}}$) and step 2 for semantic genomics imputation ($\boldsymbol{L_{\text{Task}}}$, $\boldsymbol{L_{\text{MA}}}$, and $\boldsymbol{L_{\text{SGI}}}$), with loss value after each epoch training visualized.

where $f_{i,t}$ denotes the fused prototype for the t-th matched cross-modal pair. This merge operation is parameter-free and is used to preserve interpretability while capturing shared cross-modal semantics. To retain modality-unique information, prototypes not participating in Top-K correspondences are concatenated directly without fusion. The resulting representation for each sample i thus integrates three complementary components: histology-specific, genomics-specific prototypes, and fused cross-modal prototypes. This design allows selective fusion of matched prototype pairs while retaining modality-specific features, supporting robust interpretability and fine-grained cross-modal representation learning.

E. Training Objectives and Alternating Training Strategy

The overall training objective is: $L = L_{\text{MA}} + L_{\text{SGI}} + L_{\text{Task}}$, where L_{Task} denotes standard cross-entropy loss for diagnosis and grading tasks and NLL loss [21] for survival prediction. To ensure stable convergence, we adopt an alternating training pipeline that progressively introduces different objectives, as shown in Fig. 5. The pipeline consists of two phases for each downstream task: **Step 1: Completed multimodal learning.** We first train the multimodal backbone with task-specific (e.g., classification or survival prediction) and alignment objectives, thereby enabling stable, aligned multimodal representations. Premature adversarial training is avoided to prevent disruption of fundamental feature learning. **Step 2: Semantic Genomics Imputation.** Once multimodal embeddings converge, adversarial learning in SGI is introduced to reconstruct missing genomics with a progressively increased missing rate during training, forcing the model to infer robust representations in various incomplete settings. Meanwhile, once MA achieves stable cross-modal coherence, its parameters are frozen to preserve the modal balance established. This alternating training allows each module to progressively converge and interact optimally, achieving a stable multimodal learning and generation.

IV. Experiments & Results

A. Datasets and Implementations

1) Datasets: We used two public datasets, TCGA-GBM and TCGA-LGG [46], including both WSIs and genomic profiles for gliomas. Following studies [36], [47], we removed low-quality WSIs and samples without diagnostic annotations, retaining 939 patients and 2,853 WSIs. To evaluate cross-cohort transferability, we included CPTAC-GBM [48], containing 177 glioblastoma patients with paired WSI and genomics.

2) Implementations: Our model was implemented in PyTorch and trained on NVIDIA Tesla A100 GPUs. For all downstream tasks, the model was trained for 10 epochs in TCGA-GBM/LGG cohorts [46] (in-domain). For survival prediction, we fine-tuned the pre-trained model for 10 epochs on the CPTAC-GBM [48] cohort (cross-domain) to assess transfer-learning ability. We adopted the Adam optimizer with an initial learning rate of $1e^{-4}$. We followed the standard MIL procedure, where each WSI was first divided into non-overlapping 224 $\times$ 224 pixel patches at 20$\times$ magnification (0.5 μm px^{-1}). Patch-level features were then extracted using a pre-trained CLIP ViT-B/16 backbone. The same preprocessing, including magnification, patch size, and extraction strategy, was used for all baselines to ensure a fair comparison.

B. Comparison with SOTA Methods

We evaluated performance using AUC, accuracy, sensitivity, specificity, and F1-score for glioma diagnosis and grading, and concordance index (C-index) for survival prediction. Results were obtained using patient-level five-fold cross-validation and averaged across five random splits. We compared three uni-modal methods (histology or genomic only) and twelve multimodal methods, including three classic multimodal fusion methods (Add, Concat, HFBSurv [45]), two functional gene groups-based multimodal methods (MCAT [21] and CMTA [4]), two pathway-based multimodal methods (PIBD [13] and MMP [3]), one retrieval-based missing modality method (M^2Surv [26]), one generative-based missing modality method (LD-CVAE [22]), three distillation-based missing modality methods (MKD [24], G-HANet [35], and Dispro [23]), to evaluate the effectiveness of our model.

1) Glioma Diagnosis and Grading: As shown in Table I, unimodal histology models (e.g., AttMIL [44], TransMIL [7]) perform suboptimally for both glioma diagnosis and grading, indicating that morphology alone is insufficient and that genomics provides essential complementary information. In contrast, our method, even under the missing-modality setting (only WSI inputs), outperforms the WSI-only SOTA by 2.71% and 3.79% in AUC for diagnosis and grading, respectively. In the multimodal setting, naive fusion strategies such as addition or concatenation struggle to capture cross-modal dependencies, resulting in limited performance (e.g., Concat yields F1-score 61.26% for grading). More advanced methods that explicitly model cross-modal interactions, such as HFBSurv [45], MCAT [21], CMTA [4], and MKD [24], achieve stronger results, but still fall short of our approach, demonstrating the effectiveness of our inter-modal representation strategies.

Overall, our model consistently performs the best across both diagnosis and grading. For diagnosis, our method attains the highest AUC (92.26%), surpassing the second-best MMP (90.86%) by 1.4%. For grading, our model achieves the best AUC (89.62%), outperforming MMP (88.71%) by 0.91%,

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TABLE I

COMPARISON WITH SOTA METHODS ON GLIOMA DIAGNOSIS AND GRADING. P. AND G. REPRESENT THE MODALITY OF HISTOLOGY AND GENOMICS, RESPECTIVELY. BEST AND SECOND RESULTS ARE HIGHLIGHTED WITH **BOLD** AND UNDERLINE.

Methods	p.	g.	Diagnosis (%)					Grading (%)				
			AUC	Acc.	Sen.	Spec.	F1-score	AUC	Acc.	Sen.	Spec.	F1-score
AttMIL [44]	✓		$77.26^{\pm 4.28}$	$53.14^{\pm 3.11}$	$41.53^{\pm 2.85}$	$79.32^{\pm 1.24}$	$32.96^{\pm 4.89}$	$81.47^{\pm 4.53}$	$64.29^{\pm 4.33}$	$51.72^{\pm 2.39}$	$76.65^{\pm 3.91}$	$52.86^{\pm 4.75}$
TransMIL [7]	✓		$68.28^{\pm 1.98}$	$43.50^{\pm 2.75}$	$43.60^{\pm 2.73}$	$77.06^{\pm 1.75}$	$30.14^{\pm 2.03}$	$72.88^{\pm 3.24}$	$57.40^{\pm 4.32}$	$48.09^{\pm 5.83}$	$75.37^{\pm 2.80}$	$47.94^{\pm 5.14}$
MKD [24]	✓		$82.09^{\pm 2.54}$	$60.37^{\pm 3.86}$	$53.29^{\pm 3.87}$	$82.01^{\pm 1.08}$	$47.36^{\pm 3.80}$	$83.52^{\pm 2.91}$	$66.78^{\pm 2.08}$	$\mathbf{71.68^{\pm 1.22}}$	$80.43^{\pm 2.07}$	$64.27^{\pm 3.85}$
M^2Surv [26]	✓		$83.61^{\pm 2.31}$	$65.34^{\pm 3.02}$	$57.68^{\pm 3.14}$	$84.28^{\pm 1.88}$	$52.93^{\pm 2.26}$	$83.01^{\pm 2.27}$	$67.86^{\pm 2.58}$	$66.23^{\pm 2.81}$	$79.34^{\pm 2.18}$	$63.04^{\pm 2.73}$
LD-CVAE [22]	✓		$83.83^{\pm 1.68}$	$63.17^{\pm 3.54}$	$54.76^{\pm 3.81}$	$83.20^{\pm 1.76}$	$50.93^{\pm 2.11}$	$82.94^{\pm 2.90}$	$67.15^{\pm 2.06}$	$65.88^{\pm 2.45}$	$78.47^{\pm 2.51}$	$62.29^{\pm 4.04}$
Ours (Missing Modality)	✓		$\mathbf{86.54^{\pm 2.44}}$	$\mathbf{66.30^{\pm 4.36}}$	$\mathbf{55.39^{\pm 3.45}}$	$\mathbf{85.41^{\pm 1.34}}$	$\mathbf{53.98^{\pm 5.47}}$	$\mathbf{87.31^{\pm 2.67}}$	$\mathbf{72.28^{\pm 1.67}}$	$70.35^{\pm 2.38}$	$\mathbf{82.16^{\pm 0.32}}$	$\mathbf{68.19^{\pm 3.77}}$
SNN [40]		✓	$86.29^{\pm 2.53}$	$69.13^{\pm 6.12}$	$64.60^{\pm 6.25}$	$88.36^{\pm 2.56}$	$59.53^{\pm 7.26}$	$85.70^{\pm 1.88}$	$73.02^{\pm 2.47}$	$68.19^{\pm 4.28}$	$84.25^{\pm 1.95}$	$66.30^{\pm 3.16}$
Add	✓	✓	$89.68^{\pm 4.17}$	$68.50^{\pm 2.48}$	$68.80^{\pm 2.48}$	$89.26^{\pm 0.27}$	$57.58^{\pm 2.00}$	$87.28^{\pm 2.66}$	$72.78^{\pm 4.40}$	$69.42^{\pm 3.88}$	$80.29^{\pm 1.74}$	$67.15^{\pm 4.29}$
Concat	✓	✓	$90.73^{\pm 3.06}$	$69.32^{\pm 2.13}$	$65.76^{\pm 5.11}$	$88.95^{\pm 3.92}$	$63.07^{\pm 4.57}$	$86.69^{\pm 4.52}$	$69.54^{\pm 3.74}$	$64.20^{\pm 3.29}$	$76.68^{\pm 4.02}$	$61.26^{\pm 2.34}$
HFBSurv [45]	✓	✓	$85.64^{\pm 1.72}$	$65.20^{\pm 3.18}$	$62.98^{\pm 4.53}$	$86.33^{\pm 2.54}$	$60.42^{\pm 6.14}$	$83.94^{\pm 2.46}$	$65.98^{\pm 4.14}$	$67.21^{\pm 3.28}$	$79.38^{\pm 0.96}$	$62.37^{\pm 4.53}$
MCAT [21]	✓	✓	$89.80^{\pm 1.81}$	$72.84^{\pm 3.56}$	$69.57^{\pm 3.06}$	$90.17^{\pm 1.29}$	$68.86^{\pm 2.98}$	$88.26^{\pm 1.76}$	$\mathbf{75.18^{\pm 3.06}}$	$\underline{72.44}^{\pm 5.00}$	$83.92^{\pm 7.13}$	$65.97^{\pm 3.28}$
CMTA [4]	✓	✓	$90.47^{\pm 2.39}$	$74.50^{\pm 2.41}$	$67.22^{\pm 5.03}$	$89.64^{\pm 1.61}$	$\underline{71.34}^{\pm 0.91}$	$87.80^{\pm 1.49}$	$74.74^{\pm 1.93}$	$70.24^{\pm 4.89}$	$85.02^{\pm 1.86}$	$\underline{70.25}^{\pm 3.94}$
PIBD [13]	✓	✓	$89.12^{\pm 2.03}$	$74.28^{\pm 2.37}$	$69.21^{\pm 2.18}$	$89.44^{\pm 1.57}$	$67.74^{\pm 1.95}$	$87.38^{\pm 1.92}$	$72.86^{\pm 1.81}$	$70.14^{\pm 2.03}$	$84.02^{\pm 1.96}$	$67.92^{\pm 2.08}$
MMP [3]	✓	✓	$\underline{90.86}^{\pm 1.79}$	$\underline{76.21}^{\pm 2.04}$	$\underline{71.52}^{\pm 1.88}$	$\underline{90.78}^{\pm 1.43}$	$69.83^{\pm 1.71}$	$\underline{88.71}^{\pm 1.67}$	$74.26^{\pm 1.48}$	$71.86^{\pm 1.79}$	$\underline{85.43}^{\pm 1.71}$	$69.52^{\pm 1.85}$
MKD [24]	✓	✓	$88.57^{\pm 2.76}$	$70.34^{\pm 3.06}$	$67.86^{\pm 3.20}$	$88.15^{\pm 2.74}$	$67.52^{\pm 3.23}$	$87.08^{\pm 2.19}$	$73.56^{\pm 2.81}$	$72.08^{\pm 3.16}$	$84.29^{\pm 2.56}$	$69.46^{\pm 3.23}$
M^2Surv [26]	✓	✓	$87.34^{\pm 1.95}$	$72.62^{\pm 2.45}$	$67.86^{\pm 2.07}$	$87.17^{\pm 1.52}$	$65.94^{\pm 1.83}$	$86.72^{\pm 1.79}$	$72.05^{\pm 1.72}$	$69.43^{\pm 1.92}$	$83.61^{\pm 1.76}$	$67.24^{\pm 1.88}$
LD-CVAE [22]	✓	✓	$87.96^{\pm 2.01}$	$73.39^{\pm 2.80}$	$68.57^{\pm 1.31}$	$87.82^{\pm 1.59}$	$66.86^{\pm 1.42}$	$86.15^{\pm 1.87}$	$71.03^{\pm 2.56}$	$67.02^{\pm 2.74}$	$82.50^{\pm 2.29}$	$65.15^{\pm 2.36}$
Ours	✓	✓	$\mathbf{92.26^{\pm 2.47}}$	$\mathbf{77.84^{\pm 2.31}}$	$\mathbf{73.89^{\pm 1.52}}$	$\mathbf{92.08^{\pm 1.23}}$	$\mathbf{72.25^{\pm 1.74}}$	$\mathbf{89.62^{\pm 1.37}}$	$\underline{74.80}^{\pm 1.18}$	$\mathbf{74.29^{\pm 1.56}}$	$\mathbf{87.42^{\pm 2.15}}$	$\mathbf{72.89^{\pm 1.57}}$

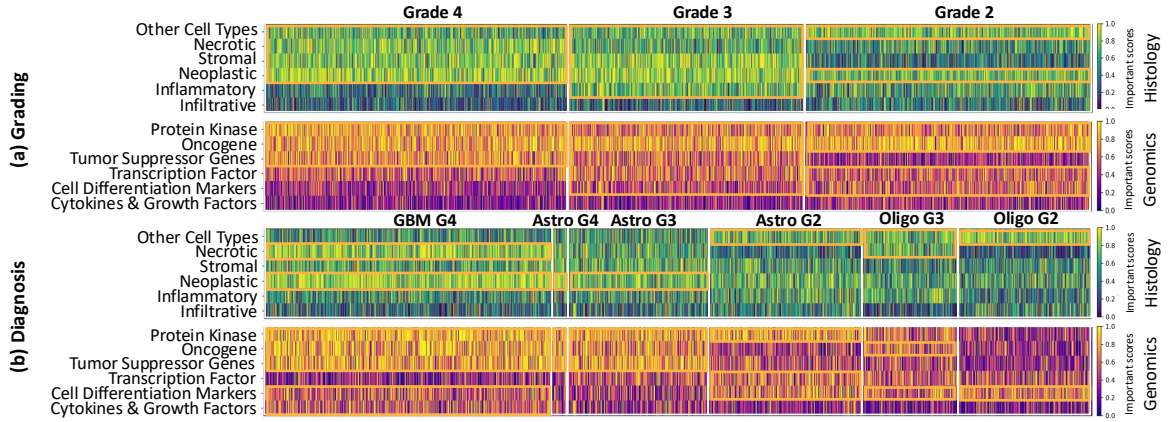


Fig. 6. Interpretability heatmaps for glioma analysis across histology and genomics. (a) Glioma Grading: patients are grouped in Grade 2-4. (b) Glioma Diagnosis (WHO 2021): GBM Grade 4, Astrocytoma Grade 4, Astrocytoma Grade 3, Astrocytoma Grade 2, Oligodendroglioma Grade 3, Oligodendroglioma Grade 2. In each task, upper heatmaps display histology prototypes (e.g., Neoplastic, Necrotic), while lower heatmaps illustrate the importance of genomic prototypes (e.g., Protein Kinase, Oncogenes). The importance score is normalized per patient. Our model shows interpretability by identifying biologically meaningful markers for glioma grading and diagnosis.

while improving F1-score by 2.64% over CMTA. These consistent improvements demonstrate the effectiveness of our intra- and inter-modal representations, and the robustness under full and missing-modality, and cross-domain conditions.

2) Survival Prediction: As shown in Table II, for survival prediction, our missing-modality model achieves the best C-index of 75.95% over other missing-modality comparisons, demonstrating its robustness for incomplete inputs. In multi-modal settings, existing methods such as MMP and M^2Surv obtain competitive in-domain results but exhibit limited cross-domain generalization. By contrast, our framework achieves the highest C-index both in-domain (83.17%) and cross-domain (64.01%) performance, highlighting improved generalization and distributional robustness of our model.

C. Robustness to Genomic Modality Missingness

To evaluate the robustness of our method under controlled genomic missingness, we simulated two settings: **Patient-wise missingness** is defined as cases where genomics is entirely unavailable for certain patients, mirroring situations where sequencing cannot be performed due to constraints of costs, tissue or resources; **Feature-wise missingness** refers to scenarios where specific genes are absent (simulated in our experiments by randomly masking subsets of genes), simulating variability in the sequencing panel or experimental noise. In both settings, missingness was simulated at predefined rates by masking the available genomic data. This controlled design retains the observed genomic profiles as references for evaluation. Together, these settings enable assessments of the impact of complex missing scenarios on model performance. We compared three representative strategies for addressing missingness using CMTA [4] as the backbone. The filling strategy replaces missing genomics with mean values. AE [49] trains an autoencoder to reconstruct genomic features from WSIs representations. The ensemble approach trains separate models on each modality and aggregates their predictions.

As shown in Table III, in the **patient-wise missingness** setting, baseline methods degrade in performance substantially as

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TABLE II

COMPARISON WITH SOTA METHODS ON SURVIVAL PREDICTION. P. AND G. REPRESENT HISTOLOGY AND GENOMICS, RESPECTIVELY.

Methods	p.	g.	Survival (C-Index, %)	
			In-domain	Cross-domain
AttMIL [44]	✓		$70.61^{\pm 3.41}$	50.37
TransMIL [7]	✓		$69.56^{\pm 3.02}$	50.84
MKD [24]	✓		$71.83^{\pm 2.16}$	49.25
G-HANet [35]	✓		$73.72^{\pm 3.24}$	50.02
M^2Surv [26]	✓		$74.06^{\pm 2.85}$	50.59
LD-CVAE [22]	✓		$72.46^{\pm 1.97}$	51.33
Dispro [23]	✓		$75.28^{\pm 2.03}$	51.47
Ours (Missing Modality)	✓		$\mathbf{75.95^{\pm 2.01}}$	**52.91**
SNN [40]		✓	$76.46^{\pm 4.32}$	<u>62.18</u>
Add	✓	✓	$78.21^{\pm 2.14}$	59.20
Concat	✓	✓	$77.10^{\pm 4.26}$	60.75
HFBSurv [45]	✓	✓	$79.67^{\pm 3.73}$	61.09
MCAT [21]	✓	✓	$81.80^{\pm 2.56}$	53.64
CMTA [4]	✓	✓	$79.24^{\pm 3.75}$	51.13
PIBD [13]	✓	✓	$80.92^{\pm 2.67}$	55.84
MMP [3]	✓	✓	$82.74^{\pm 2.18}$	58.63
MKD [24]	✓	✓	$80.76^{\pm 3.59}$	54.67
M^2Surv [26]	✓	✓	<u>$82.89^{\pm 1.82}$</u>	60.79
LD-CVAE [22]	✓	✓	$81.09^{\pm 1.84}$	53.27
Dispro [23]	✓	✓	$82.18^{\pm 2.06}$	60.02
Ours	✓	✓	$\mathbf{83.17^{\pm 2.29}}$	**64.01**

TABLE III

PERFORMANCE COMPARISON UNDER DIFFERENT TYPES OF GENOMIC MISSINGNESS ON SURVIVAL PREDICTION. BEST AND SECOND RESULTS ARE HIGHLIGHTED WITH **BOLD** AND UNDERLINE.

Type	Methods	C-Index (%) at different missing rates				
		100%	80%	50%	20%	0%
Patient-wise	Filling	$58.38^{\pm 5.76}$	$65.38^{\pm 3.10}$	$70.21^{\pm 4.61}$	$74.90^{\pm 5.29}$	<u>$79.24^{\pm 3.75}$</u>
	AE	$61.16^{\pm 5.63}$	$65.60^{\pm 3.17}$	$72.20^{\pm 4.56}$	<u>$76.94^{\pm 2.76}$</u>	<u>$79.24^{\pm 3.75}$</u>
	Ensemble	<u>$69.43^{\pm 4.89}$</u>	<u>$70.96^{\pm 5.20}$</u>	<u>$75.38^{\pm 2.14}$</u>	$76.01^{\pm 4.19}$	$79.02^{\pm 3.02}$
	Ours	$\mathbf{75.95^{\pm 2.01}}$	$\mathbf{76.04^{\pm 3.10}}$	$\mathbf{77.82^{\pm 2.35}}$	$\mathbf{79.38^{\pm 3.22}}$	$\mathbf{83.17^{\pm 2.29}}$
Feature-wise	Filling	$58.38^{\pm 5.76}$	$64.84^{\pm 3.63}$	$74.48^{\pm 3.57}$	$78.90^{\pm 2.28}$	<u>$79.24^{\pm 3.75}$</u>
	AE	$61.16^{\pm 5.63}$	$69.36^{\pm 5.14}$	$74.03^{\pm 3.59}$	$78.92^{\pm 3.10}$	<u>$79.24^{\pm 3.75}$</u>
	Ensemble	<u>$69.43^{\pm 4.89}$</u>	<u>$70.35^{\pm 2.37}$</u>	<u>$74.91^{\pm 5.28}$</u>	<u>$79.15^{\pm 3.24}$</u>	$79.02^{\pm 3.02}$
	Ours	$\mathbf{75.95^{\pm 2.01}}$	$\mathbf{75.94^{\pm 4.30}}$	$\mathbf{77.74^{\pm 3.43}}$	$\mathbf{81.38^{\pm 2.95}}$	$\mathbf{83.17^{\pm 2.29}}$

the proportion of missingness increases. The filling approach with mean values struggles to maintain predictive accuracy, as it fails to capture biological heterogeneity. The AE approach is limited in genomic representations due to substantial multi-modal semantic gaps. The ensemble approach is comparatively more stable by leveraging multiple modality-specific models; however, it fails to model cross-modal interactions, resulting in suboptimal performance. Our method consistently outperforms these baselines, as the proposed SGI module effectively reconstructs genomic embeddings from histology-derived features, ensuring biological coherence. These advantages are particularly evident at a higher missing ratio.

A similar pattern is observed in feature-wise missingness, demonstrating the flexibility and robustness of our method. Compared to patient-wise missingness, the performance decline in **feature-wise** is generally more gradual, as models can still utilize the remaining genomics. Nonetheless, as the missing rate increases, the accumulation of noise adversely affects both filling and AE methods, resulting in performance degradation. Meanwhile, the ensemble approach becomes sensitive to missingness. In contrast, our method achieves robust performance on all feature-wise missingness. Additionally, on a WSI-only TCGA-GBM/LGG [46] cohort of 295 patients without available genomic profiles, our method achieved a C-index of 68.3% without fine-tuning, outperforming strong competing methods M^2Surv (65.4%) and LD-CVAE (67.1%).

Collectively, these findings highlight our advantages in maintaining biologically coherent representations, preserving cross-modal consistency, and robustly imputing incomplete genomics. The strong performance across diverse levels of missingness demonstrates the robustness of our framework to different types of simulated and real genomic missingness.

D. Ablation Study

As shown in Table IV, we conducted comprehensive ablation studies of four proposed modules on all tasks under both missing and complete genomics.

1) Ablation of BP module: We compared four initialization strategies: random initialization, CLIP-based text prompts, PLIP-based simple text prompts, and ours (PLIP-based, more specific text prompts initialization). All text-prompt-based methods outperform random initialization, indicating the effectiveness of biologically grounded prototype initialization. PLIP-based prompts initialization is superior to CLIP-based prompts initialization, indicating that pathology foundation models may provide a more generalized semantic representation. We also found that text prompts that provide more specific pathological descriptions (ours) outperform the ones with simple descriptions (ours with simple descriptions), suggesting that more specific morphological cues could help unlock the potential of histology images.

2) Ablation of MA module: We separately removed the sample-wise and distribution-wise alignment components. Ablating either degrades performance, while removing both leads to the most pronounced degradation. The sample-wise alignment improves robustness across both full and missing modality settings, while the distribution-wise alignment offers additional improvements, particularly when genomics are missing, as it regularizes latent feature dependencies and guides imputation. These findings confirm that the two alignment mechanisms are complementary and jointly crucial for robust cross-modal representation learning.

3) Ablation of SGI module: Removing the SGI module forces the model to rely solely on paired WSI-genomic data, disabling its ability to handle missing modalities. A diffusion-based imputation variant underperforms, likely due to its limited capacity to model biologically meaningful cross-modal dependencies. Similarly, a GAN-based variant without the proposed progressive interpolation degrades sharply under partial missingness, underscoring the importance of gradually exposing incomplete genomics during training. Moreover, even with all modalities available at inference, incorporating SGI during training consistently enhances performance. This can be attributed to the regularization effect of adversarial and reconstruction objectives, which strengthen cross-modal consistency and promote biologically coherent latent representations.

4) Ablation of BF module: Replacing BF with a simple concatenation or standard cross-attention results in clear performance degradation, particularly under missing modality

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TABLE IV

ABLATION STUDY OF MODEL COMPONENTS UNDER MISSING AND FULL GENOMIC CONDITIONS. BEST RESULTS ARE HIGHLIGHTED WITH **BOLD**.

Module	Variant	Missing Modality (%)			Full Modality (%)		
		Diagnosis (Acc.)	Grading (Acc.)	Survival (C-Index)	Diagnosis (Acc.)	Grading (Acc.)	Survival (C-Index)
BP	w/o BP (Random Init)	$63.57^{\pm 4.51}$	$69.05^{\pm 2.93}$	$74.19^{\pm 5.96}$	$75.01^{\pm 0.19}$	$70.58^{\pm 2.81}$	$80.17^{\pm 3.77}$
	Ours (CLIP Init)	$65.84^{\pm 3.22}$	$70.86^{\pm 2.63}$	$74.63^{\pm 2.34}$	$75.32^{\pm 1.18}$	$71.68^{\pm 2.46}$	$80.76^{\pm 2.57}$
	Ours (PLIP simple description Init)	$65.21^{\pm 4.05}$	$71.03^{\pm 2.71}$	$74.45^{\pm 2.16}$	$76.01^{\pm 2.24}$	$72.72^{\pm 1.58}$	$82.41^{\pm 2.38}$
	Ours	$\mathbf{67.52^{\pm 3.94}}$	$\mathbf{72.30^{\pm 2.84}}$	$\mathbf{75.95^{\pm 2.01}}$	$\mathbf{77.84^{\pm 2.31}}$	$\mathbf{74.83^{\pm 2.67}}$	$\mathbf{83.17^{\pm 2.29}}$
MA	w/o MA	$45.69^{\pm 5.48}$	$58.42^{\pm 2.92}$	$65.48^{\pm 3.19}$	$65.48^{\pm 3.19}$	$66.27^{\pm 2.64}$	$78.33^{\pm 4.56}$
	w/o Sample-wise Alignment	$62.42^{\pm 4.08}$	$69.88^{\pm 2.43}$	$72.15^{\pm 2.52}$	$73.58^{\pm 3.98}$	$72.96^{\pm 1.84}$	$80.59^{\pm 3.92}$
	w/o Distribution-wise Alignment	$63.59^{\pm 3.97}$	$69.72^{\pm 3.07}$	$69.55^{\pm 4.18}$	$74.62^{\pm 2.73}$	$71.50^{\pm 2.87}$	$81.36^{\pm 5.02}$
	Ours	$\mathbf{67.52^{\pm 3.94}}$	$\mathbf{72.30^{\pm 2.84}}$	$\mathbf{75.95^{\pm 2.01}}$	$\mathbf{77.84^{\pm 2.31}}$	$\mathbf{74.83^{\pm 2.67}}$	$\mathbf{83.17^{\pm 2.29}}$
SGI	w/o SGI	—	—	—	$76.28^{\pm 1.81}$	$73.25^{\pm 2.19}$	$82.02^{\pm 2.39}$
	Diffusion-based Imputation	$48.71^{\pm 5.28}$	$50.47^{\pm 3.84}$	$60.33^{\pm 6.29}$	$76.28^{\pm 1.81}$	$73.25^{\pm 2.19}$	$82.02^{\pm 2.39}$
	GAN-based (w/o Interpolation)	$60.85^{\pm 3.83}$	$69.46^{\pm 2.53}$	$72.06^{\pm 3.30}$	$76.80^{\pm 3.82}$	$73.02^{\pm 1.47}$	$82.93^{\pm 3.10}$
	Ours	$\mathbf{67.52^{\pm 3.94}}$	$\mathbf{72.30^{\pm 2.84}}$	$\mathbf{75.95^{\pm 2.01}}$	$\mathbf{77.84^{\pm 2.31}}$	$\mathbf{74.83^{\pm 2.67}}$	$\mathbf{83.17^{\pm 2.29}}$
BF	w/o BF (Replace with Concatenation)	$63.52^{\pm 5.46}$	$70.33^{\pm 2.56}$	$73.42^{\pm 3.67}$	$74.09^{\pm 2.83}$	$72.56^{\pm 2.43}$	$82.56^{\pm 2.00}$
	Replace with Cross Attention	$65.38^{\pm 4.62}$	$71.22^{\pm 2.33}$	$73.92^{\pm 2.45}$	$76.96^{\pm 2.02}$	$73.52^{\pm 1.97}$	$82.83^{\pm 2.08}$
	Ours	$\mathbf{67.52^{\pm 3.94}}$	$\mathbf{72.30^{\pm 2.84}}$	$\mathbf{75.95^{\pm 2.01}}$	$\mathbf{77.84^{\pm 2.31}}$	$\mathbf{74.83^{\pm 2.67}}$	$\mathbf{83.17^{\pm 2.29}}$

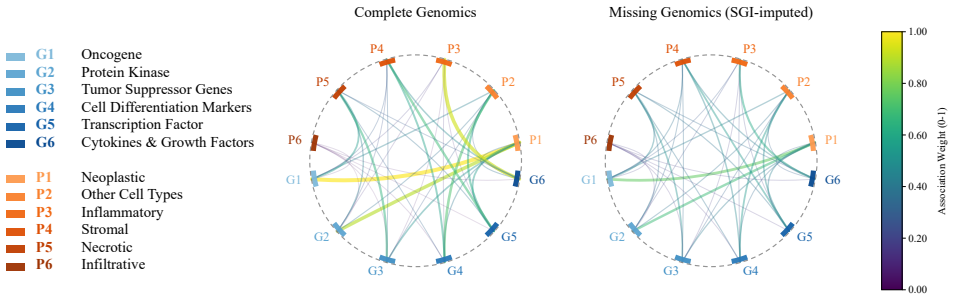


Fig. 7. Visualization of cross-modal interactions between genomic prototypes (G1-G6) and histology prototypes (P1-P6) on the survival prediction task, with complete genomics and imputed genomics by SGI. Edge color represents geno-phenotype association weights.

TABLE V

HYPERPARAMETER ANALYSIS OF $\boldsymbol{\lambda}$ IN THE SGI MODULE FOR SURVIVAL PREDICTION (C-INDEX, %).

Setting	$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
Full modality ↑	82.93	**83.19**	83.17	81.32	79.01
Missing modality ↑	62.45	72.56	**75.95**	70.34	60.08

TABLE VI

HYPER-PARAMETER ANALYSIS ON THE TOP-$\boldsymbol{K}$ PARAMETER IN THE BF MODULE FOR SURVIVAL PREDICTION. (C-INDEX, %)

Metric	$K = 0$	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$	$K = 6$
In-domain ↑	82.56	82.43	83.09	**83.17**	82.91	82.62	82.35
Cross-domain ↑	63.19	63.22	**64.88**	64.01	63.73	63.44	63.17
Train time (h) ↓	3.83	3.77	3.71	3.67	3.62	3.56	3.53

settings. This may be due to its ability to suppress noisy or redundant features arising from imputation, while retaining informative modality-specific signals. This selective fusion proves more effective than naive or classic fusion, underlining its importance in managing heterogeneous multimodal data.

E. Analysis of Hyper-Parameters

1) Sensitivity to λ in SGI module: We further analyzed the sensitivity of the SGI objective in Eq. 10 to the weighting parameter λ, which balances adversarial learning and cycle consistency. We evaluated $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ under both full- and missing-modality settings. As shown in Table V, the model achieves the best overall trade-off at $\lambda = 0.5$, which yields the strongest performance under missing modality and remains near-optimal under full modality. In contrast, extreme values of λ led to degraded performance, especially when genomics was missing. The results indicate that moderate weighting between adversarial loss and cycle consistency is important for robust semantic genomics imputation.

2) Sensitivity to K in BF module: We further investigated the effect of the Top-K parameter in the BF module on survival prediction. As described in Section III-D, Top-K controls how many cross-modal prototype pairs are fused, balancing integration and modality-specific preservation. Table VI shows that performance remains relatively stable across different K. However, smaller K values (e.g., $K = 0, 1$) underexploit cross-modal interactions, resulting in weaker performance in both in-domain and cross-domain settings. The setting $K = 3$ achieves the highest in-domain accuracy (83.17%), while also providing competitive cross-domain performance (64.01%), only slightly below the best cross-domain result obtained at $K = 2$ (64.88%). In terms of computational efficiency, both $K = 2$ and $K = 3$ reduce training time compared to smaller values, likely because fusing similar pairs suppresses redundant information and accelerates convergence. Overall, $K = 3$ provides the best balance between in-domain performance,

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Fig. 8. T-SNE visualization of histology and genomic features before and after multimodal alignment over 10 training epochs across three downstream tasks: diagnosis, grading, and survival prediction. Blue and orange points denote histology and genomics embeddings, respectively.

generalization and training efficiency, and is thus adopted as the default configuration.

F. Interpretability Analysis

1) Prototype-level Interpretability: To investigate biological patterns learned by our model, we visualize multimodal importance contributions across patients in Fig. 6. Each column represents a patient, grouped by clinical labels, and rows refer to proposed prototypes derived from WSIs or genomics. Color intensity denotes relative importance scores of each prototype.

In Fig. 6 (a), Grade 4 gliomas show high reliance on oncogenes and protein kinases in genomic prototypes, consistent with high-grade tumor hallmarks such as cell proliferation and immune escape [50]. Also, their histological profiles emphasize neoplastic and necrotic components, characteristics of aggressive tumors. Grade 3 exhibits an intermediate dependence on transcription factors and tumor suppressor genes, accompanied by a greater emphasis on inflammatory histology. Grade 2 is characterized by cell differentiation marker genes and relatively normal-appearing tissue in histology, indicating a more indolent phenotype.

Fig. 6 (b) further shows that the model differentiates glioma subtypes by capturing subtype-specific genomic and histology signatures. For example, GBM (Grade 4, IDH-wildtype) prominently emphasizes oncogenic signaling and necrotic morphology, whereas astrocytomas exhibit a grade-dependent transition from prominent oncogene signaling (in Grades 4 and 3) to increased involvement of transcriptional regulation and differentiation (in Grade 2). Oligodendrogliomas, particularly in lower grades, demonstrate consistent importance of cell differentiation markers and histology of other cell types. Overall, the model-derived prototype importance patterns show qualitative correspondence with prior knowledge and reasonable cross-seed consistency, with mean pairwise Spearman correlations of 0.83 and 0.82 for diagnosis and grading, respectively.

2) Cross-modal Prototypes Interaction: Fig. 7 visualizes the learned cross-modal interactions for the survival prediction task using a chord diagram under complete genomics and missing genomics reconstructed by SGI. Genomic prototypes (G1-G6) and histology prototypes (P1-P6) are represented as nodes, with edges reflecting the magnitude of their associations. Biological associations can be observed: P1 (Neoplastic) exhibits strong links with G1 (Oncogene) and G2 (Protein Kinase), consistent with the role of oncogenic signaling and kinase activity in tumor proliferation. Similarly, P3 (Inflammatory) shows strong connections with G6 (Cytokines & Growth Factors), reflecting the involvement of cytokine-mediated signaling in shaping tumor microenvironment and immune responses. Overall association patterns are partially preserved after imputation, with P1-G1 remaining the strongest association under SGI-imputed genomics. These model-derived associations show qualitative consistency with established cancer biology [50], [51], although further validation is still needed before drawing biological conclusions from them.

3) Multimodal Feature Alignment: To illustrate the effect of the MA module, Fig. 8 visualizes the evolution of multimodal representations during training. At the initial epoch, histology and genomics embeddings are completely separated, indicating weak cross-modal correspondence. After applying our MA module, the two modalities become progressively intertwined, revealing improved feature coherence. By mid-training (e.g., Epoch 5), embeddings after MA exhibit strong alignment across modalities, suggesting that our MA module effectively bridges the semantic gap between histology and genomics. Toward later epochs, both the *before* and *after* alignment representations appear well aligned, implying that the alignment loss has been internalized by the encoders through end-to-end optimization. This progression highlights MA as both an early-stage calibrator and a continuing regularizer, promoting consistent multimodal representations across tasks.

4) WSI-level Interpretability: Fig. 9 depicts the spatial attention heatmaps of six histology prototypes on WSIs in survival prediction. Each prototype exhibits distinct spatial distributions. The *Neoplastic* prototype shows strong activation in tumor-dense regions, while the *Stromal* prototypes are more active in stromal areas. The *Inflammatory* prototype highlights immune infiltration, and the *Necrotic* prototype aligns with necrotic cores commonly observed in high-grade gliomas. These spatially coherent activation patterns indicate that the model effectively captures biologically meaningful regions, aligning with known histopathological features. The attention maps also show reasonable cross-seed stability, with a mean Top-100 high-attention patch overlap of 0.69 across histology prototypes.

5) Failure Cases: Despite strong overall performance, we observed several failure modes. **First**, the learned histology prototypes do not always localize their corresponding tissue regions precisely. This is particularly evident for histological concepts where boundaries are ambiguous on H&E alone. Even experts may require additional immunohistochemical staining or ancillary context. **Second**, the necrotic prototype

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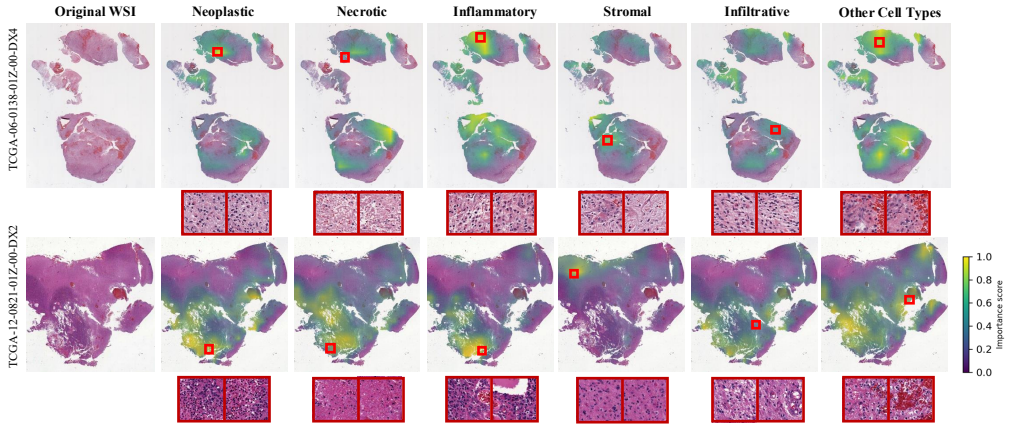


Fig. 9. Spatial attention heatmaps of six histology prototypes on WSIs, showing phenotype-specific activation patterns for survival prediction.

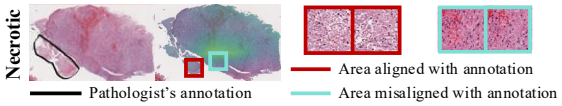


Fig. 10. Failure case of necrotic prototype attention. The model partially matches the annotated necrotic region but also highlights blood-rich non-necrotic tissue, indicating confusion with hemorrhagic morphology under weak supervision.

highlights not only pathologist-identified necrotic regions but also blood-rich areas as shown in Fig. 10. A possible reason is that both regions may share similar morphological cues, such as reduced nuclear density and disrupted tissue structure.

V. Conclusion

This study presents an interpretable multimodal prototyping framework that integrates histology and genomic data for precision oncology. Our approach jointly addresses four key challenges in multimodal learning: suboptimal WSI representation, ineffective cross-modal alignment, coarse integration, and limited robustness with incomplete genomics.

Through BP module, the framework embeds domain priors into representation learning, enhancing both clinical relevance and model transparency; the MA strategy enforces semantic consistency through sample-wise and distribution-wise alignment, calibrating the imperfect pairing in multimodal dataset; the BF module captures both shared and complementary signals via cross-modal affinity; and the SGI module reconstructs genomic semantics from histology, ensuring robustness under incomplete or entire missing genomic scenarios. Extensive experiments across diagnosis, grading, and survival prediction demonstrate that our framework not only surpasses state-of-the-art models but also remains resilient when genomics are absent. These results highlight that biologically-informed prototyping, semantic alignment, and selective fusion enhance interpretability, robustness, and clinical reliability.

Our work has limitations. First, more in-depth analysis of genomics and histology could benefit biological interpretation and relevance. Future efforts will incorporate pathway-level modeling to capture biological interactions. Meanwhile, from the failure cases, we found that even BP module enhances histology interpretability to some extent, our model may fail in some complex cases, as the intrinsic ambiguity of H&E morphology. Future work may benefit from more fine-grained pathology prompts, more advanced pathology foundation models, and genomic pathways to better separate biologically similar but diagnostically distinct tissue patterns. In addition, the attribution visualizations lack quantitative region-level localization and validation by multiple pathologists. Future work will evaluate their pathological validity using dedicated expert annotations and, where necessary, matched IHC staining. Second, our controlled missingness experiments were simulated from complete cases and therefore do not fully capture panel-specific, institution-dependent, or non-random missingness in clinical practice. Further validation on cohorts with more complex missingness will be necessary before clinical deployment. Future work will extend to multi-institutional and pan-cancer datasets to enhance clinical translation. Also, although this study focuses on histology and genomics, our framework is readily extensible to additional modalities, such as demographic data, through late-fusion strategies. Overall, this study represents a step towards developing interpretable and robust multimodal learning methods under missing modalities for precision oncology.

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